

(54) CRYPTOGRAPHIC COMMUNICATIONS SYSTEM AND METHOD

(57) Abstract

A cryptographic system based on the difficulty of factoring the product of two large primes. Public and private exponents are derived such that decryption inverts encryption modulo n .

Detailed Description

Let p and q be distinct primes and let $n = p \cdot q$. Define $\phi(n) = (p-1)(q-1)$. Choose e coprime to $\phi(n)$ and let d be the multiplicative inverse of e modulo $\phi(n)$:

$$e \cdot d \equiv 1 \pmod{\phi(n)}$$

Encryption of message $m \in \mathbb{Z}_n$ proceeds as $c = m^e \bmod n$, and decryption as $m = c^d \bmod n$. Correctness follows from Euler's theorem:

$$m^{(e \cdot d)} \equiv m^{(1 + k \cdot \phi(n))} \equiv m \pmod{n}$$

Claims

1. A communications method comprising: encoding a plaintext m as an integer in $[0, n)$; computing $c = m^e \bmod n$; transmitting c ; and recovering $m = c^d \bmod n$ at a receiver, wherein $n = p \cdot q$ for distinct primes p, q and $d \equiv e^{-1} \bmod (p-1)(q-1)$.
2. The method of claim 1, wherein $|p| = |q| = 1024$ bits.
3. The method of claim 1, wherein $e = 65537$.